



Hybrid Advantage Shaping with Goal-Aware Attention for Per-Turn Credit Assignment in LLM-Agent Reinforcement Learning

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Motivation

LLM agents acting over many turns in sparse-reward environments such as AlfWorld and WebShop receive only a single terminal reward, yet success depends on credit being assigned to the individual turns that actually mattered. As tool-use, web, and embodied agents tackle ever longer horizons, stable per-turn credit assignment has become the central bottleneck for sample-efficient learning.

Existing approaches leave this gap open: trajectory-level GRPO (DeepSeekMath) copies one advantage onto every token and gives no per-turn signal; step-level GRPO (GiGPO, HoGPO) adds per-step structure but relies on fixed, heuristic credit; and return decomposition (RUDDER-style) learns per-step credit yet remains goal-agnostic. We therefore propose:

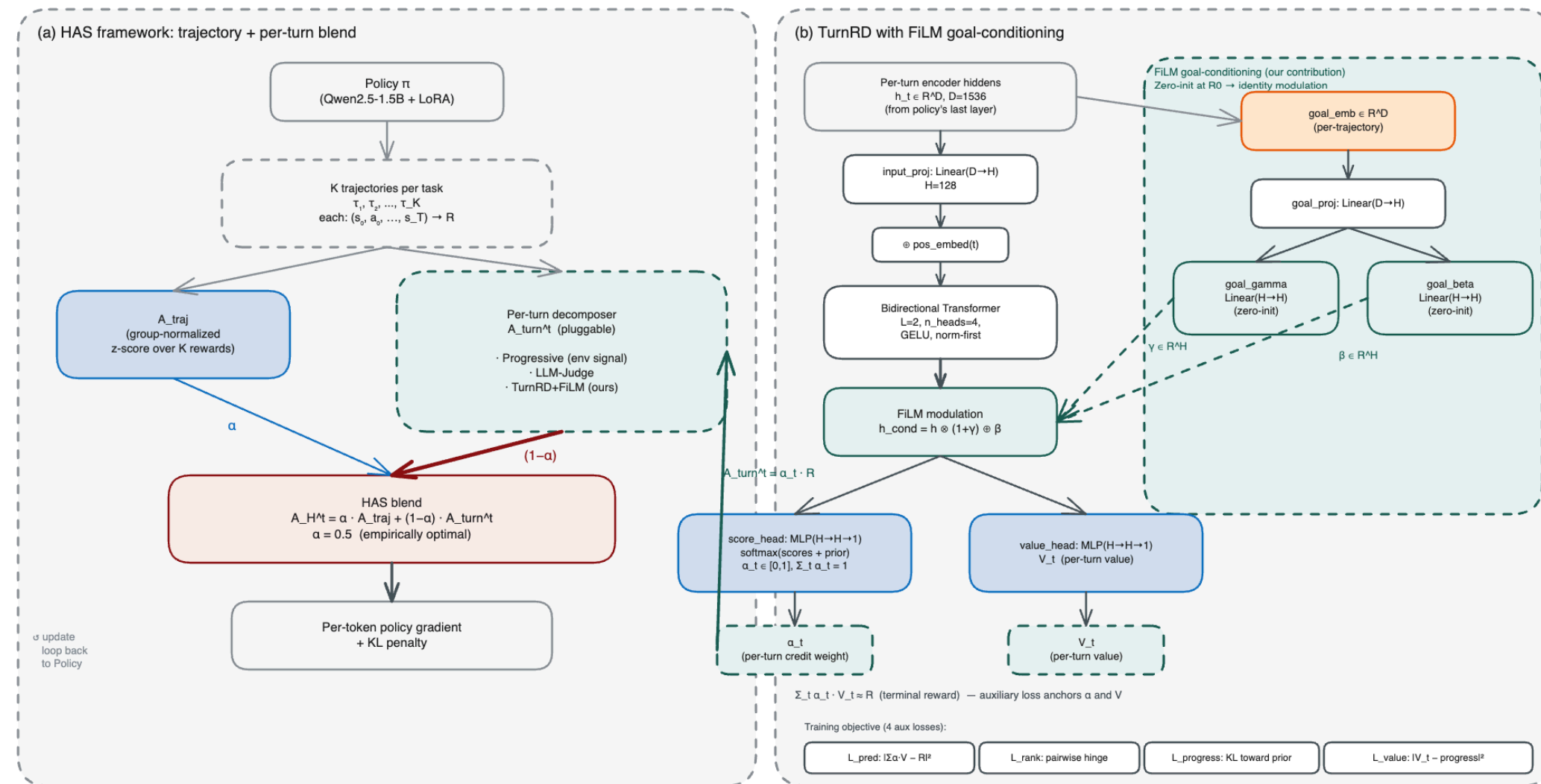
- **Hybrid Advantage Shaping (HAS)** — a one-knob framework that blends a trajectory baseline with a pluggable per-turn decomposer.
- **TurnRD with goal-aware FiLM attention** — the strongest decomposer: learnable attention over per-turn hiddens, modulated by a per-trajectory goal embedding.

Method: Hybrid Advantage Shaping

Per-turn advantage is a blend of the trajectory-level baseline and a per-turn signal:

$$A_H^t = \alpha \cdot A_{\text{traj}} + (1 - \alpha) \cdot A_{\text{turn}}^t$$

- $\alpha = 1.0$: pure trajectory (flatGRPO, no credit); $\alpha = 0.5$: hybrid blend (empirically optimal)
- Per-turn decomposer is pluggable: Progressive, LLM-Judge, or TurnRD (learned attention)



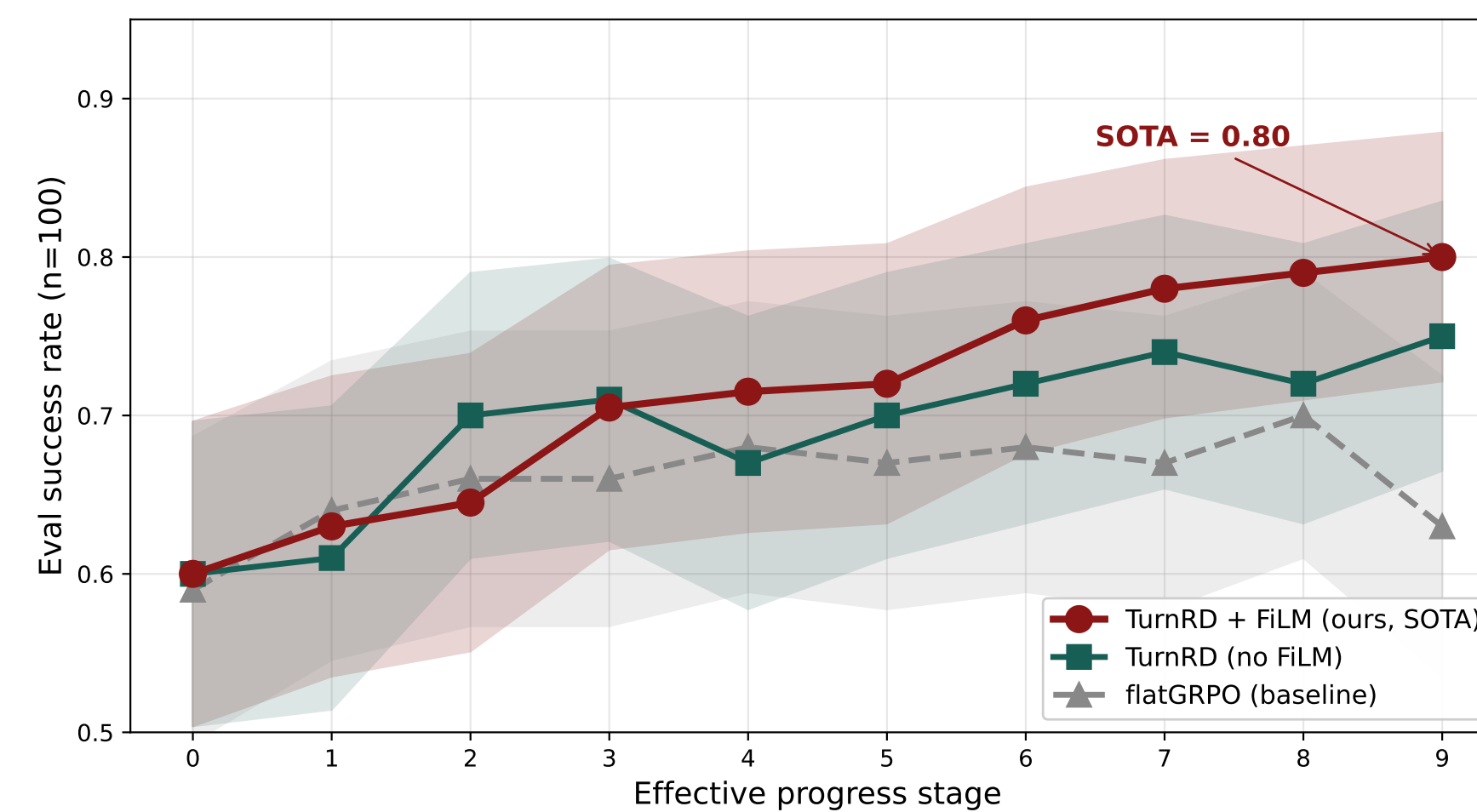
TurnRD + FiLM (goal aware)

- **Input/Encoder:** per-turn hiddens h_t (policy's last hidden state), encoded by a bidirectional transformer (2 layers, 4 heads, $H = 128$)
- **FiLM (Feature-wise Linear Modulation) goal-conditioning:** per-trajectory goal_emb produces γ, β ; $\text{FiLM}(h_t) = \gamma \odot h_t + \beta$ modulates h_t before α - and V -heads
- **α -head:** softmax over turns gives the per-turn credit weight α_t
- **V -head:** per-turn value V_t ; $\sum_t \alpha_t V_t \approx R$ recovers reward
- **Training:** 4-term auxiliary loss (pred, rank, progress, value); re-fit per round on cumulative recency-decayed replay buffer

Experiment & Results

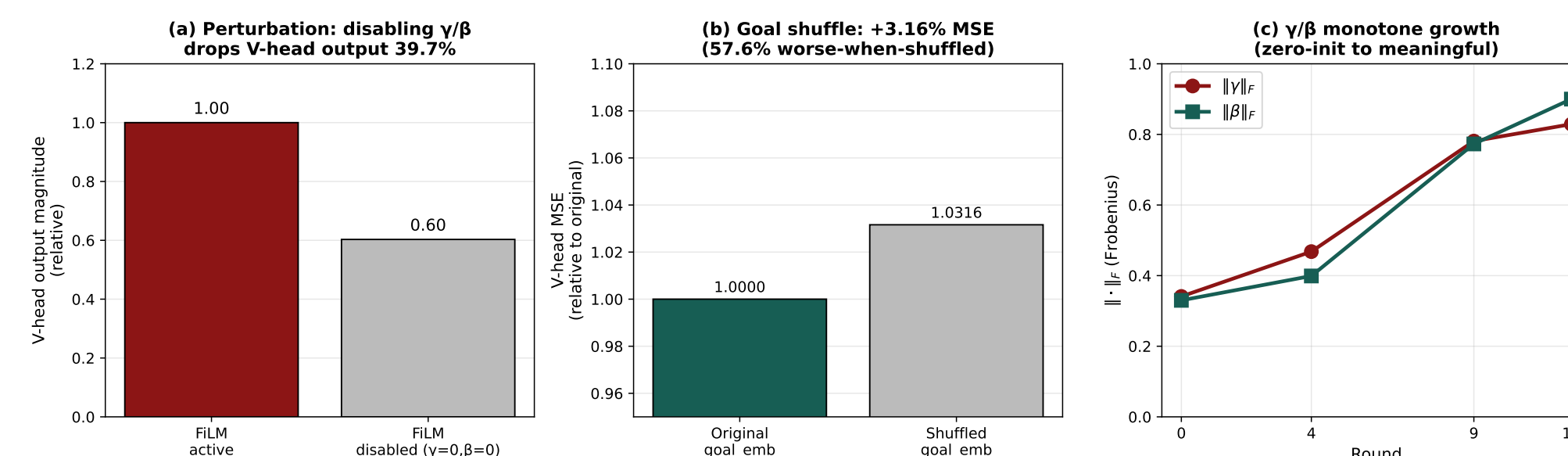
- **Models:** Qwen2.5-1.5B-Instruct + LoRA $r = 32$ (attn + MLP targets)
- **Envs:** AlfWorld (1,000 games), WebShop (1,000 products)
- **Training:** 10 rounds $\times$ 80 episodes $\times$ $K = 8$ rollouts/task
- **Eval:** greedy $K = 1$, $n = 100$ held-out episodes per round
- **Compute:** A100-80GB on Modal, ~ 5 –12h per run
- **AlfWorld:** 80% success vs $\sim 70\%$ flatGRPO baseline (+10pp)
- **WebShop:** 91.3% success vs 80.5% flatGRPO (+10.8pp)

AlfWorld — TurnRD+FiLM reaches 80%



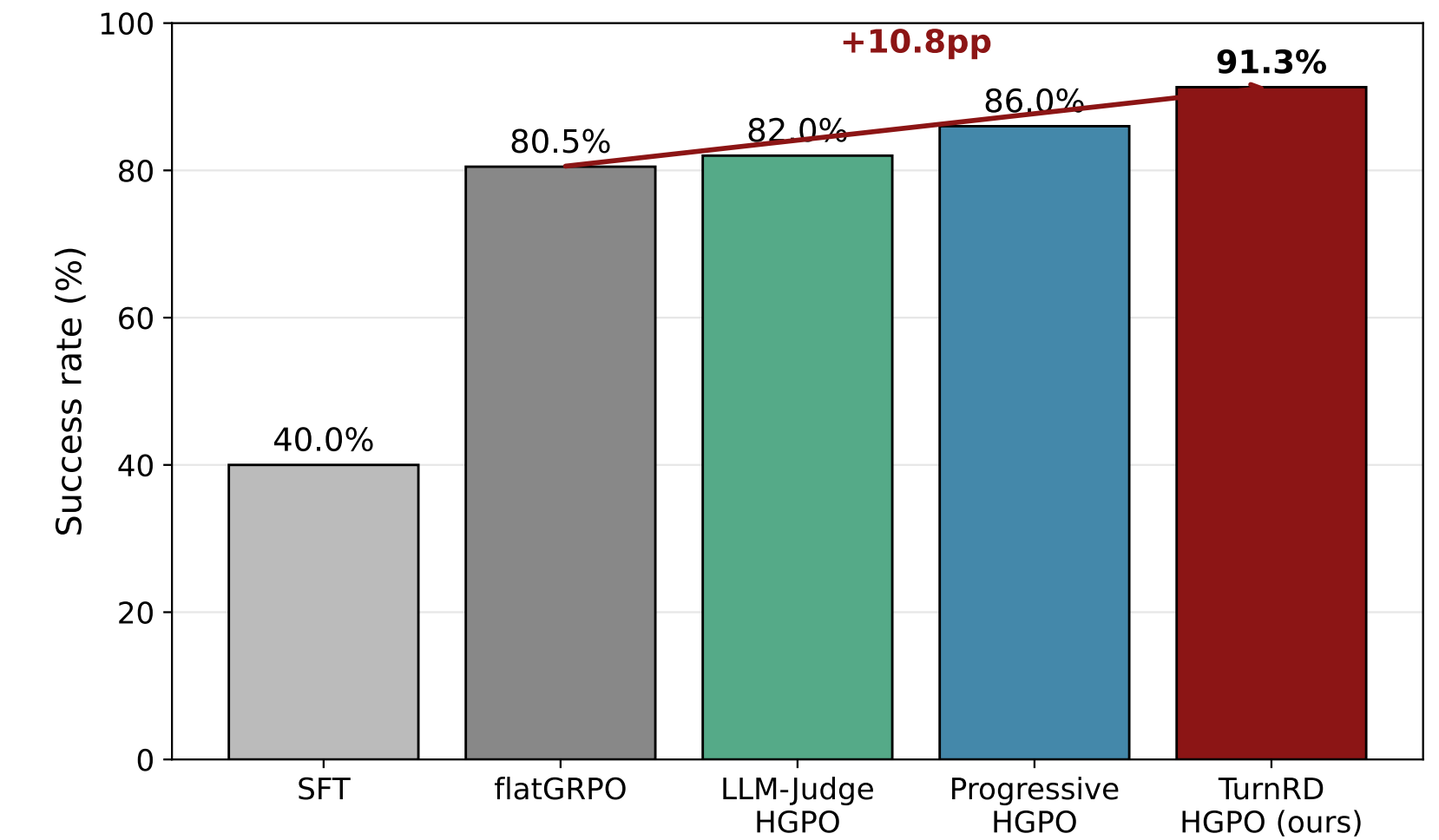
- TurnRD+FiLM (80%) > TurnRD without FiLM (75%) > flatGRPO ($\sim 70\%$): the per-turn signal and goal-aware FiLM each add measurable lift.
- Our method improves steadily to 80%, while flatGRPO plateaus in the high-60s%.
- Early rounds are noisy and overlap the baseline; TurnRD's lead emerges with more training as its learned credit assignment matures.

Goal-Aware FiLM is Actively Learning



- **Perturbation:** zeroing γ/β changes the per-turn V-head credit by 39.7%, so goal conditioning genuinely shapes the assigned credit.
- **Goal-shuffle:** feeding a mismatched goal_emb worsens V-head MSE by 3.16% (57.6% of samples), so the credit tracks the true goal, not just position.
- **γ/β growth:** modulation strength rises monotonically (γ norm 0.34 to 0.83, R0–R12), so the model leans more on the goal as training proceeds.

WebShop — HAS generalizes across envs



- The hybrid blend ($\alpha = 0.5$) beats the flatGRPO baseline ($\alpha = 1.0$) by 10.8pp, so the framework generalizes beyond AlfWorld.
- TurnRD's learned attention beats Progressive (+5.3pp) and LLM-Judge (+9.3pp): learned per-turn credit outperforms hand-designed heuristics.

Conclusion

- **One knob, general:** HAS blends trajectory- and turn-level advantages with a single α and improves results in both environments.
- **Learned, goal-aware credit wins:** TurnRD with FiLM is the strongest decomposer, adding +5pp on AlfWorld over the no-FiLM variant.
- **Strong end-to-end results:** 80% on AlfWorld and 91.3% on WebShop from one shared framework.
- **Safe by design:** at $\alpha = 1.0$ HAS reduces to flatGRPO, so it is never worse than the baseline.

Limitations & Future Work

- **Single base model:** validated only on Qwen2.5-1.5B with LoRA; full fine-tuning and larger backbones remain untested.
- **Limited environments:** results cover AlfWorld and WebShop only; transfer to other agentic domains is unverified.
- **Modest goal signal:** the goal-shuffle effect (+3.16%) is statistically real but small, so goal-awareness is present yet weak.
- **Future work:** adaptive per-round α schedules, larger models, multi-task transfer, and a stronger goal-discrimination loss.

References

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- [3] Zhihong Shao et al. DeepSeekMath: Pushing the limits of mathematical reasoning in open language models, 2024. Preprint.
- [4] Ethan Perez et al. FiLM: Visual reasoning with a general conditioning layer. In AAAI, 2018.